

MATRIX REPRESENTATIONS OF OPERATORS ON FINITE-DIMENSIONAL INNER PRODUCT SPACES

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ABSTRACT. This is a short article that proves a useful result involving the matrix representation of an operator on a finite-dimensional inner-product space.

1. INTRODUCTION

The structure provided by an inner-product space enables operators on that space to be neatly expressed in matrix form. This is useful in quantum mechanics to determine the matrix representation of the angular momentum operators.

Recall that the matrix of an operator T over a finite-dimensional vector space V with respect to a basis $v_1, \dots, v_n$ is the object

$$\mathcal{M}(T, (v_1, \dots, v_n)) = \begin{pmatrix} A_{1,1} & \dots & A_{1,n} \\ \vdots & & \vdots \\ A_{n,1} & \dots & A_{n,n} \end{pmatrix}$$

whose entries¹ $A_{j,k} \in \mathbb{F}$ are determined by the action of T on a basis vector v_k :

$$Tv_k = \sum_{j=1}^n A_{j,k} v_j.$$

These entries may be tedious to calculate, as one would need to apply T to each v_k . Over inner-product spaces, however, the matrix of T is much easier to determine.

Theorem 1. *Let V be a finite-dimensional inner-product space, equipped with an orthonormal basis² $e_1, \dots, e_n$. Then for any operator $T \in \mathcal{L}(V)$, the matrix of T with respect to this basis is*

$$(1) \quad \mathcal{M}(T, (e_1, \dots, e_n)) = \begin{pmatrix} \langle Te_1, e_1 \rangle & \dots & \langle Te_n, e_1 \rangle \\ \vdots & & \vdots \\ \langle Te_1, e_n \rangle & \dots & \langle Te_n, e_n \rangle \end{pmatrix}.$$

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¹Here, $\mathbb{F}$ denotes the scalar field; either $\mathbb{F} = \mathbb{C}$ or $\mathbb{F} = \mathbb{R}$.

²That V admits an orthonormal basis is guaranteed by the Gram-Schmidt Algorithm applied to a (not necessarily orthonormal) basis of V , which must exist since V is assumed to be finite-dimensional.

The result above may be proved in two different ways; we discuss both of them.

Proof 1: Let the matrix of T with respect to $e_1, \dots, e_n$ be

$$\mathcal{M}(T, (e_1, \dots, e_n)) = \begin{pmatrix} A_{1,1} & \dots & A_{1,n} \\ \vdots & & \vdots \\ A_{n,1} & \dots & A_{n,n} \end{pmatrix}.$$

For each Te_k , we have, by definition,

$$Te_k = \sum_{r=1}^n A_{r,k} e_r.$$

Then, for a vector basis vector e_j , where $j \in \mathbb{Z} \cap [1, n]$,

$$\begin{aligned} \langle Te_k, e_j \rangle &= \left\langle \sum_{r=1}^n A_{r,k} e_r, e_j \right\rangle \\ &= \sum_{r=1}^n A_{r,k} \langle e_r, e_j \rangle \\ &= \sum_{r=1}^n A_{r,k} \delta_{r,j} \\ &= A_{j,k}, \end{aligned}$$

where $\delta_{r,j}$ denotes the standard Kronecker delta; the third equality holds by orthonormality. This completes the proof. ■

Proof 2: Alternatively, one can denote $v_k = Te_k$, and observe ³

$$v_k = \sum_{r=1}^n A_{r,k} e_r = \sum_{m=1}^n \langle v_k, e_m \rangle e_m.$$

Unique representation implies that $A_{j,k} = \langle v_k, e_j \rangle = \langle Tv_k, e_j \rangle$. As k was an arbitrary integer between 1 and n , we have our result. ■

This result can be used to derive a neat formula for the trace of an operator over an inner-product space V over $\mathbb{F}$; for an orthonormal basis $e_1, \dots, e_n$ of V , the diagonal entries of $\mathcal{M}(T, (e_1, \dots, e_n))$ are given by $\langle Te_k, e_k \rangle$ by above, so the definition of the trace implies⁴

$$\text{Tr } T = \sum_{k=1}^n \langle Te_k, e_k \rangle.$$

The result of Theorem 1 has a nice corollary:

³That v can be expressed in such a manner is 6.30(a) in [Axl24].

⁴Note that the trace of a matrix of an operator is independent of the chosen basis— see 8.54 in [Axl24].

Theorem 2. *Let V be a finite-dimensional ($\dim V = n < \infty$) inner-product space, and suppose $T \in \mathcal{L}(V)$ is a self-adjoint operator. Then all eigenvalues of T are real-valued.*

Proof: The appropriate Spectral Theorem⁵ implies that there exists a orthonormal basis $e_1, \dots, e_n$ of V with respect to which the matrix of T is diagonal. The diagonal entries, which by Theorem 1 are given by $\langle Te_k, e_k \rangle$, are precisely the eigenvalues of T . As T is self-adjoint, we have

$$\langle Te_k, e_k \rangle = \langle e_k, Te_k \rangle = \overline{\langle Te_k, e_k \rangle}.$$

The equation above shows that each $\langle Te_k, e_k \rangle$ is real, as desired. ■.

We can obviously extend our result to a function $T \in \mathcal{L}(V, W)$, where V and W are finite-dimensional inner product spaces. In this case, if $e_1, \dots, e_n$ and $f_1, \dots, f_m$ are orthonormal bases of V and W , the entries of $\mathcal{M}(T, (e_1, \dots, e_n), (f_1, \dots, f_m))$ are given by $A_{j,k} = \langle Te_k, f_j \rangle$, where, of course, the inner product is taken over the target space W .

It is handy to be able to express the matrix of an operator in the clean manner displayed in Equation 1, especially in quantum mechanics. For example, the matrix of the square of the spin vector $\hat{S} = (\hat{S}_x, \hat{S}_y, \hat{S}_z)$ for a spin $\frac{1}{2}$ particle can be written⁶

$$\mathcal{M}\left(\hat{S}^2, \left\{\left|\frac{1}{2}, m_s\right\rangle\right\}_{-\frac{1}{2} \leq m_s \leq \frac{1}{2}}\right) = \begin{pmatrix} \langle \frac{1}{2}, \frac{1}{2} | \hat{S}^2 | \frac{1}{2}, \frac{1}{2} \rangle & \langle \frac{1}{2}, \frac{1}{2} | \hat{S}^2 | \frac{1}{2}, -\frac{1}{2} \rangle \\ \langle \frac{1}{2}, -\frac{1}{2} | \hat{S}^2 | \frac{1}{2}, \frac{1}{2} \rangle & \langle \frac{1}{2}, -\frac{1}{2} | \hat{S}^2 | \frac{1}{2}, -\frac{1}{2} \rangle \end{pmatrix}$$

which reduces neatly to the diagonal matrix

$$\frac{3\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Further calculation of this sort (on $\hat{S}_z$ and the ladder operators) eventually lead to the derivation of the Pauli matrices for a particle of spin $\frac{1}{2}$, which are useful in describing the behaviour of, say, an electron in an electromagnetic field via the Schrödinger-Pauli Equation. For further details, consult chapters 5 and 6 of [Zet22].

REFERENCES

- [Axl24] Sheldon Axler. *Linear Algebra Done Right*. Undergraduate Texts in Mathematics. Springer, 2024. ISBN: 978-3031410253.
- [Zet22] Nouredine Zettili. *Quantum Mechanics*. Wiley, 2022. ISBN: 978-1-118-30789-2.

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⁵This depends on the choice of the scalar field $\mathbb{F}$. Note the Complex Spectral Theorem holds for normal operators, which is a weaker assumption than that for the Real Spectral Theorem.

⁶Note that the convention of the inner-product in quantum mechanics is different to that in pure math; physicists take the inner-product to be conjugate linear in the *first* entry. Do note, however, that the spin (and for that matter general angular momentum) operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$ and $\hat{S}^2$ are self-adjoint (Hermitian).